main.py

Visualize Run

```python
import pandas as pd

df = pd.DataFrame({'ID':[101,102,103,104,105],
'Name':['Ravi','Kumar','Anu','Priya','Manoj'],
'Department':['IT','HR','IT','HR','Sales'],
'Salary':[50000,45000,65000,55000,60000]})

print(df.loc[df.groupby('Department')['Salary'].idxmax(),
                ['Department','Name','Salary']].sort_values('Departmen
```

Output

```
  Department   Name  Salary
3         HR  Priya   55000
2         IT    Anu   65000
4      Sales  Manoj   60000

=== Code Execution Successful ===
```

main.py

Visualize Run

```python
import pandas as pd

df = pd.DataFrame({'Product':['Laptop','Mobile','Mouse','Keyboard',
'Tablet'],
'Category':['Electronics','Electronics','Accessories','Accessories',
'Electronics'],
'Rating':[4.5,4.2,3.8,4.6,4.8]})

avg = df.groupby('Category')['Rating'].mean()
print(avg.round(2))
print('Highest Rated Category:', avg.idxmax(), round(avg.max(),2))
```

Output

```
Category
Accessories    4.2
Electronics    4.5
Name: Rating, dtype: float64
Highest Rated Category: Electronics 4.5

=== Code Execution Successful ===
```

main.py

Visualize Run

```python
import pandas as pd

df = pd.DataFrame({'Name':['Arun','Bala','Charan','Divya'],
'Total_Days':[50,50,50,50],'Present_Days':[45,38,48,35]})

df['Attendance %'] = df.Present_Days / df.Total_Days * 100
print(df[['Name','Attendance %']])
print(df.loc[df['Attendance %'] < 80, ['Name','Attendance %']])
```

Output

```
    Product  Total_Sales
0    Laptop       250000
1     Mouse        10000
2  Keyboard        10000
3    Mobile       160000
4    Tablet        90000
Category
Accessories     20000
Electronics    500000
Name: Total_Sales, dtype: int64

=== Code Execution Successful ===
```

main.py

Visualize Run

```python
import pandas as pd

df = pd.DataFrame({'Product':['Laptop','Mouse','Keyboard','Mobile',
'Tablet'],
'Category':['Electronics','Accessories','Accessories','Electronics',
'Electronics'],
'Quantity':[5,20,10,8,6],'Price':[50000,500,1000,20000,15000]})

df['Total_Sales'] = df.Quantity * df.Price
print(df[['Product','Total_Sales']])
print(df.groupby('Category')['Total_Sales'].sum())
```

Output

```
    Product  Total_Sales
0    Laptop       250000
1     Mouse        10000
2  Keyboard        10000
3    Mobile       160000
4    Tablet        90000
Category
Accessories     20000
Electronics    500000
Name: Total_Sales, dtype: int64

=== Code Execution Successful ===
```

main.py

Visualize Run

```python
import pandas as pd

df = pd.DataFrame({'Name':['Arun','Bala','Charan','Divya'],
'Python':[80,65,90,75],'Java':[85,70,88,80],'DBMS':[75,60,92,78]})

df['Total'] = df[['Python','Java','DBMS']].sum(axis=1)
df['Average'] = df[['Python','Java','DBMS']].mean(axis=1)
print(df[['Name','Total','Average']].round(2))
print(df.loc[df.Average > 75, ['Name','Average']].round(2))
```

Output

```
     Name  Total  Average
0    Arun    240    80.00
1    Bala    195    65.00
2  Charan    270    90.00
3   Divya    233    77.67
     Name  Average
0    Arun    80.00
2  Charan    90.00
3   Divya    77.67

=== Code Execution Successful ===
```